CST546 - VLSI Design

CST 546 (VLSI Design) Syllabus

Course Information

Course Details

- **Course Title:** VLSI Design
- **Course Number:** CST 546
- **Meeting Location:** Physical
- **Units:** 2

Course Description

This course introduces students to the principles of Very Large-Scale Integration (VLSI) design, focusing on the design and development of digital systems from the transistor level up. Key topics include transistor-level design, logic gates, and the use of Field-Programmable Gate Arrays (FPGAs) for prototyping and development. Students will gain practical experience in designing and simulating VLSI circuits, with an emphasis on the real-world applications of VLSI in computer hardware design.

Materials

- Course textbook: "CMOS VLSI Design: A Circuits and Systems Perspective" by Neil H.E. Weste and David Money Harris

Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system - FPGA development boards for circuit prototyping and implementation - CAD software for VLSI circuit design and simulation (Cadence, Synopsys, or open-source alternatives) - Access to electronic design automation (EDA) tools for layout and verification

Course Objectives

At the end of this class, you should be able to:

1. Design digital systems at the transistor level using VLSI design principles
2. Implement and analyze logic gates and complex digital circuits
3. Utilize FPGAs for prototyping and developing digital system designs
4. Apply VLSI design techniques to real-world computer hardware applications

Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

Assignment Kind	Value
Projects	45%
Exams	25%
Participation	30%

Projects

Projects will take the form of basic digital circuit design exercises, simple logic gate implementations, and introductory FPGA programming assignments. Students will focus on fundamental VLSI concepts through simulation tools and straightforward hardware design challenges.

Exams

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

Participation

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups. Therefore, attendance is required, as well as participation in all in-class activities. These activities may take the form of in-class discussions, brainstorming sessions, stand-ups, as well as making active progress on projects during class time.

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.